

Using Multilevel QD Projective Method for Solving Multigroup Neutron Transport Coupled with Heat Transfer

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Project Overview

Multilevel Nonlinear Projective method projects multigroup transport onto one-group Quasidiffusion (QD) equation for scalar flux.

- QD equation coupled to TH via implicit temperature dependence of cross sections
- Coupled system of GLOQD and TH solved with Newton iteration

Project scope:

- Implement MNP method for coupled neutron transport and TH
 - Linear Discontinuous Finite Element method for Transport Equation, MLOQD, and GLOQD
 - Finite volume method for advection-diffusion TH equation
- Fourier stability analysis for MNP method

Compare to NILO-CMFD method

Outline

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- 3 Multilevel Low-Order Quasidiffusion
- 4 Project Scope

Problem Definition I

$$\mathcal{L}(T)\psi(\vec{r}, \hat{\Omega}, E, t) = \mathcal{S}(T)\psi(\vec{r}, \hat{\Omega}, E, t) + q(\vec{r}, \hat{\Omega}, E, t) + \sum_k \mathcal{P}_k C_k, \quad (\text{Neutron Transport})$$

$$\frac{\partial C_k}{\partial t} = -\lambda_k C_k + \int_0^\infty dE \beta_k(E) \nu(E) \Sigma_f(E) \int_{4\pi} d\hat{\Omega} \psi(\vec{r}, \hat{\Omega}, E, t), \quad (\text{Precursor Balance})$$

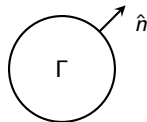
$$\mathcal{H}T(\vec{r}) = \mathcal{Q}(T)\psi(\vec{r}, \hat{\Omega}, E, t), \quad (\text{Heat Transfer})$$

$$\Sigma_u(\vec{r}) = \Sigma_u(T, \phi) \quad (\text{XS Temp. Dependence})$$

$$\psi(\vec{r}, \Omega, E, 0) = \psi^0, \quad (1d)$$

$$T(\vec{r}, 0) = T^0, \quad (1e)$$

$$\psi(\vec{r}, \Omega, E, t) = \psi^{in}, \quad \hat{n} \cdot \hat{\Omega} < 0, \vec{r} \in \partial\Gamma \quad (1f)$$



Problem Definition II

$$\mathcal{L}(T)\psi = \left[\frac{\partial}{\partial t} + \mu \frac{\partial}{\partial z} + \Sigma_t(T) \right] \psi \quad (2a)$$

$$\begin{aligned} \mathcal{S}(T)\psi = & \frac{\chi(E)}{2} \int_0^\infty dE' \nu \Sigma_f(E', T) \int \phi(z, E', t) \\ & + \frac{1}{2} \int_0^\infty dE' \Sigma_s(E' \rightarrow E, T) \phi(z, E', t) \end{aligned} \quad (2b)$$

$$\Sigma_i = \Sigma_{i,0} + \Sigma_{i,1,d} \left(\sqrt{T + \beta_{fb} W_f f(z)} - \sqrt{T_{d,0}} \right) + \Sigma_{i,1,b} (T - T_0) \quad (2c)$$

$$\mathcal{P}_k C_k = \frac{1}{2} \lambda_k \chi_k^{del}(z, E, t) C_k(z, t) \quad (2d)$$

$$\mathcal{Q}(T)\psi = \int_0^\infty dE W_f(E) \Sigma_f(E, T) \phi(z, E, t) \quad (2e)$$

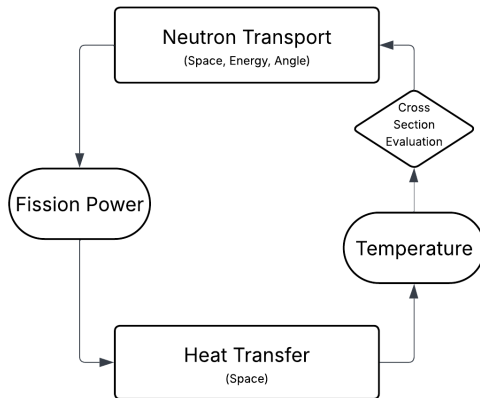
$$\mathcal{H}T = \left[\frac{\partial}{\partial t} + \dot{m} c_p \frac{\partial}{\partial z} - \frac{\partial}{\partial z} k_b \frac{\partial}{\partial z} \right] T(z) \quad (2f)$$

FPI Coupling

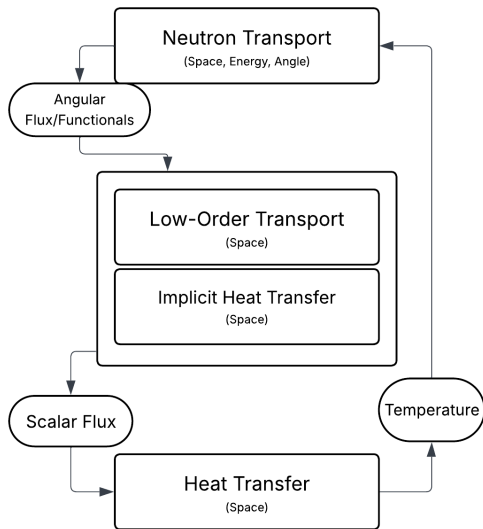
Directly HT to High-Order Transport is expensive

Transport: $\vec{r}, \hat{\Omega}, E, t$

Heat Transfer: $\vec{r}, t$



Coupling with Low-Order Operators



- Coupling is contained in LO/Approximate system
- Requires few solutions of NT and HT equations
- Can approximate TH in inner iterations without approximation in final solution

CMFD Coupling Methods – FPI-CMFD

CMFD aids iteration by introducing Diffusion Synthetic Acceleration (DSA) and coarse-mesh finite difference (CMFD).

Most apparent coupling method: fixed-point iteration (FPI), which treats cross sections *explicitly*

$$\boxed{\Sigma_i(T^n)} \xrightarrow{\substack{\uparrow \\ \mathcal{D} \text{ from transport sweep}}} \boxed{\mathcal{D}\phi^{(n+1)} = \mathcal{S}_0} \longrightarrow \boxed{\mathcal{H}T^{(n+1)} = \mathcal{Q}\phi^{(n+1)}} \quad (\text{FPI})$$

CMFD-FPI is an Ineffective method

- Slow to converge
- May be unstable
- Solution of HT equation is expensive

CMFD Coupling Methods – R-CMFD

Instabilities from FPI can be resolved by “relaxing” the heat source on each iteration:

$$\boxed{\Sigma_i(T^n)} \longrightarrow \boxed{\mathcal{D}\phi^{(n+1)} = \mathcal{S}_0} \longrightarrow \boxed{\mathcal{H}T^{(n+1)} = \mathcal{Q}(\alpha\phi^{(n+1)} + (1-\alpha)\phi^{(n)})}$$

(R-CMFD)

R-CMFD still has its problems

- Slower to converge than FPI – requires more HT evaluations
- Still may be unstable

CMFD Coupling Methods – X-CMFD

Use implicit cross sections in low-order diffusion calculation

- Diffusion uses end-of-iteration xs

$$\boxed{\mathcal{L}\psi^{(n+1)} = \mathcal{S}\phi^{(n)}} \longrightarrow \boxed{\begin{array}{l} \mathcal{D}(T^{(n+1)})\phi^{(n+1)} = \mathcal{S}_0 \\ \mathcal{H}T^{(n+1)} = \mathcal{Q}\phi^{(n+1)} \end{array}} \quad (\text{X-CMFD})$$

Solved by “inner iteration”

X-CMFD addresses stability

- Stable, converges in same number of outer iterations as single-physics transport
- Requires repeated (expensive) solution of HT

CMFD Coupling Methods – NILO-CMFD [1]

Reduce cost of implicit solution of coupled low-order diffusion and HT

- Use approximation $H \approx \mathcal{H}$ of HT operator to update implicit cross sections within inner iterations
- Update cross sections by Taylor expansion (rather than table lookup or function evaluation)

Implicit cross section update performed during inner iterations:

$$\Sigma^{(n+1)} = \Sigma(T^{(n+1)}, h^{(n+1)}) \quad (3)$$

$$\approx \Sigma_i^{(n)} + \frac{\partial \Sigma^{(n)}}{\partial T} \left(T^{(n+1)} - T^{(n)} \right) + \frac{\partial \Sigma}{\partial h} \left(h^{(n+1)} - h^{(n)} \right) \quad (4)$$

$$\begin{aligned} \approx \Sigma_i^{(n)} + \frac{\partial \Sigma^{(n)}}{\partial T} H^{-1} W \left(\Sigma_{f\phi}^{(n+1)} - \Sigma_{f\phi}^{(n)} \right) \\ + \frac{\partial \Sigma^{(n)}}{\partial h} W \left(\Sigma_{f\phi}^{(n+1)} - \Sigma_{f\phi}^{(n)} \right) \end{aligned} \quad (5)$$

Multigroup Low-Order Quasidiffusion

Take zeroth and first angular moments of the high-order multigroup NTE in 1D slab geometry

$$\begin{aligned} \frac{1}{v_g} \frac{\partial \phi_g}{\partial t} + \frac{\partial J_g}{\partial x} + (\Sigma_{t,g} - \Sigma_{s,g \rightarrow g}) \phi_g \\ = \sum_{g'=1}^{g-1} \Sigma_{s,g' \rightarrow g} \phi_{g'} + \left(Q_{s,g,0} + \chi_g^p \overline{(1 - \beta) \nu \Sigma_f} \right) \phi + q_{del,g}^{\chi} \end{aligned} \quad (6a)$$

$$\frac{1}{v_g} \frac{\partial J_g}{\partial t} + \frac{\partial}{\partial x} (E_g \phi_g) + \Sigma_{t,g} J_g = 0 \quad (6b)$$

$$\phi_g(x, 0) = \phi_g^0, \quad J_g(0, x) = J_g^0 \quad (6c)$$

$$J_g(0, t) = C_{0,g} \phi_g(0, t), \quad J_g(X, t) = C_{X,g} \phi_g(X, t) \quad (6d)$$

Closure terms

QD equations and high-order equations closed by Eddington and boundary factors:

$$E_g = \frac{\int_{-1}^1 d\mu \mu^2 \psi_g}{\int_{-1}^1 d\mu \psi_g}, \quad (7)$$

$$B_{0,g} = \frac{\int_{-1}^0 d\mu \mu \psi_g(0, \mu)}{\int_{-1}^0 d\mu \psi_g(0, \mu)}, \quad B_{X,g} = \frac{\int_0^1 d\mu \mu \psi_g(X, \mu)}{\int_0^1 d\mu \psi_g(X, \mu)} \quad (8)$$

Grey Low-Order Quasidiffusion (GLOQD)

Sum the Multigroup LOQD equations to yield

$$\frac{\partial}{\partial t} \left(\frac{\phi}{\bar{v}} \right) + \frac{\partial J}{\partial x} + \left(\bar{\Sigma}_a + \overline{(1 - \beta)\nu \Sigma_f} \right) \phi = q_{del}^x \quad (9a)$$

$$\frac{\partial}{\partial t} \left(\frac{J}{\bar{v}} \right) + \frac{\partial}{\partial x} (\bar{E}\phi) + \bar{\Sigma}_{tr} J + \bar{\zeta} \phi = 0 \quad (9b)$$

$$\phi(x, 0) = \phi^0, \quad J(x, 0) = J^0, \quad (9c)$$

$$J(0, t) = \bar{B}_0 \phi(0, t), \quad J(X, t) = \bar{B}_X \phi(X, t) \quad (9d)$$

Effect of projection

High-order NTE $\longrightarrow$ MLOQD $\longrightarrow$ GLOQD

$(x, \mu, E, t) \longrightarrow (x, E, t) \longrightarrow (x, t)$

Project NTE to same dimensionality as HT

General Project Goals

Combine Advantages of NILO-CMFD [1] and MLOQD [2], write my own implementation

- Couple NTE and HT using Nonlinearly Implicit Low-Order coupling and Multigroup Low-Order Quasidiffusion
- Consider delayed neutrons (not included in CMFD results)
- 1D slab geometry, Linear Discontinuous (LD) finite element method
- HT: advection-diffusion equation using finite volume method

Perform Fourier stability analysis of NILO-MLOQD method

- Ensure stability for all parameters
- Estimate performance of method (iterations saved)
- Compare Fourier results with numerical results

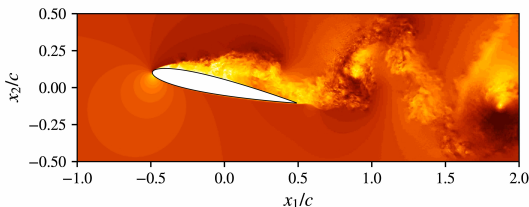
Time, spatial, and angular discretization of ELOT problem

Consider 1D slab/channel geometry with isotropic scattering. Discretize transport using

- Angle – Discrete Ordinates
- Time – Backward Euler
- Space – LD Galerkin Method

and discretize HT using Finite Volume method.

I will implement in Python, but can be scaled up to 2/3D using tools such as MFEM [3]



MFEM Navier Minapp (Spieser 2023)

References I

- [1] N. J. Adamowicz, A. Manera, and E. W. Larsen, “The NILO-CMFD method for iteratively solving coupled neutron transport–thermal hydraulics problems,” *Nuclear Science and Engineering*, vol. 197, no. 2, pp. 262–278, 2023.
- [2] A. Tamang and D. Y. Anistratov, “A multilevel projective method for solving the space-time multigroup neutron kinetics equations coupled with the heat transfer equation,” *Nuclear Science and Engineering*, vol. 177, no. 1, pp. 1–18, 2014.
- [3] R. Anderson, J. Andrej, A. Barker, J. Bramwell, J.-S. Camier, J. Cervený, V. Dobrev, Y. Dudouit, A. Fisher, T. Kolev, *et al.*, “Mfem: A modular finite element methods library,” *Computers & Mathematics with Applications*, vol. 81, pp. 42–74, 2021.